

Critical RPC Availability Issue & Network Access Bottleneck

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Environment	WSL Ubuntu (Linux CLI) — Indonesia
Report Type	Infrastructure Diagnostics — RPC Availability Investigation

Executive Summary

This report documents a network-level issue affecting DAC Testnet accessibility, identified through independent infrastructure testing and diagnostics. Multiple users reported being unable to connect via wallet or dApps, remaining stuck on "Still connecting to DAC Chain Testnet...". Testing was conducted across all enodes listed in the official source at enodes.dachain.tech/testnet.

Layer	Status	Detail
P2P / Consensus Layer	Operational	All peers reachable on port 28657
RPC Layer — Node Endpoints	Severely Limited	1 of 7 nodes responsive
Public RPC — <code>rpctest.dachain.tech</code>	Unavailable	504 Gateway Time-out (nginx)

Observed User Issue

Multiple users reported being unable to connect to the DAC Testnet via wallet clients or dApps. The common symptom across all affected users was a persistent connecting state with no resolution:

```
"Still connecting to DAC Chain Testnet..."
```

This report was initiated to identify whether the issue was network-wide or isolated, and to determine the root cause layer.

Network-Level Evidence — Official Enode Validation

All findings below are based on enodes listed in the official source at enodes.dachain.tech/testnet. Testing was conducted across the full available enode set using direct JSON-RPC probing with a 3-second timeout per endpoint.

P2P Layer

All tested nodes are reachable at the peer level (port 28657 open). This confirms the network is operational at the consensus layer — block production and peer discovery are functioning normally.

RPC Layer — Critical Findings

Out of 7 official nodes tested, only 1 node responded to JSON-RPC queries. The remaining 6 returned timeout or no response:

Node IP	RPC Status	Response
206.189.127.204	Responsive	{"result": "0xe1749e"}
157.173.127.30	No Response	Timeout (3s)
156.67.104.212	No Response	Timeout (3s)
144.91.74.181	No Response	Timeout (3s)
157.173.127.21	No Response	Timeout (3s)
157.173.127.31	No Response	Timeout (3s)
161.97.89.27	No Response	Timeout (3s)

Public RPC Endpoint

The official public RPC endpoint was tested independently:

```
Endpoint: rpctest.dachain.tech
Test:     HTTP GET / POST
Result:   504 Gateway Time-out (nginx)
```

The 504 response indicates the nginx gateway is reachable but the backend RPC nodes are not responding within an acceptable time window. This makes the public endpoint effectively unavailable to all wallet clients and dApps that rely on it.

Testing Environment

Platform	WSL Ubuntu (Linux CLI)
Region	Indonesia
Method	Direct JSON-RPC probing via curl — eth_blockNumber
Timeout	3 seconds per endpoint
Source	enodes.dachain.tech/testnet — official enode list

Impact

- Users are unable to connect to the DAC Testnet via wallets or dApps.
- Wallet clients remain stuck in a persistent "connecting" state with no resolution.
- Developer and community experience is significantly degraded.

- All activity dependent on RPC access — transactions, balance checks, contract interaction — is effectively blocked.

Conclusion

Layer	Status
Consensus / P2P Layer	Operational — block production and peer connectivity confirmed
RPC Layer	Severely limited — 6 of 7 nodes unresponsive
Public RPC Endpoint	Unavailable — 504 Gateway Time-out

The network is technically running at the consensus level, but not accessible to end users. This creates a bottleneck where the RPC layer — the interface between users and the chain — is effectively non-functional.

Recommendations

- Deploy additional RPC nodes with public access to eliminate the single-point-of-failure.
- Ensure all official nodes listed in `enodes.dachain.tech` are fully synced and have RPC ports open.
- Implement load balancing across multiple RPC backends to distribute request load.
- Provide alternative RPC endpoints for community usage as a fallback.
- Monitor RPC health metrics continuously — latency, timeout rate, and sync status.

Supporting Evidence — Batch RPC Test Output

Raw output from direct JSON-RPC probing of each official node (abridged):

```
==== RPC Test 157.173.127.30:8545 ====
No response / timeout

==== RPC Test 156.67.104.212:8545 ====
No response / timeout

==== RPC Test 144.91.74.181:8545 ====
No response / timeout

==== RPC Test 157.173.127.21:8545 ====
No response / timeout

==== RPC Test 157.173.127.31:8545 ====
No response / timeout

==== RPC Test 206.189.127.204:8545 ====
{"jsonrpc":"2.0","id":1,"result":"0xe1749e"}

==== RPC Test 161.97.89.27:8545 ====
No response / timeout
```

```
jeruzzalet@EdLWEISS:~$ while read enode; do ip_port=$(echo $enode | sed -E 's|.*([^\s]+)|\1|'); ip=$(echo $ip_port | cut -d: -f1); port=$(echo $ip_port | cut -d: -f2); echo "====  
Checking $ip:$port =====  
nc -vz -w 2 $ip $port >/dev/null 2>&1 && echo "P2P: OPEN" || echo "P2P: CLOSED";  
for rpc_port in 8545 26657; do timeout 2 bash -c "</dev/tcp/$ip/$rpc_port" >/dev/null 2>&1 && echo "RPC $rpc_port: OPEN" || echo "RPC $rpc_port: CLOSED"; done; echo  
==== Checking 157.173.127.30:28657 =====  
P2P: OPEN  
RPC 8545: CLOSED  
RPC 26657: CLOSED  
  
==== Checking 156.67.104.212:28657 =====  
P2P: OPEN  
RPC 8545: CLOSED  
RPC 26657: CLOSED  
  
==== Checking 144.91.74.181:28657 =====  
P2P: OPEN  
RPC 8545: CLOSED  
RPC 26657: CLOSED  
  
==== Checking 157.173.127.21:28657 =====  
P2P: OPEN  
RPC 8545: CLOSED  
RPC 26657: CLOSED  
  
==== Checking 157.173.127.31:28657 =====  
P2P: OPEN  
RPC 8545: CLOSED  
RPC 26657: CLOSED  
  
==== Checking 206.189.127.204:28657 =====  
P2P: OPEN  
RPC 8545: OPEN  
RPC 26657: CLOSED  
  
==== Checking 161.97.89.27:28657 =====  
P2P: OPEN  
RPC 8545: CLOSED  
RPC 26657: CLOSED  
  
jeruzzalet@EdLWEISS:~$ |
```

```
jeruzzalet@EdLWEISS:~$ nano enodes.txt  
jeruzzalet@EdLWEISS:~$ while read enode; do ip_port=$(echo $enode | sed -E 's|.*([^\s]+)|\1|'); ip=$(echo $ip_port | cut -d: -f1); echo "==== RPC Test  
t $ip:8545 ====="; curl -s --max-time 3 -X POST http://$ip:8545 -H "Content-Type: application/json" -d '{"jsonrpc":"2.0","method":"eth_blockNumber","p  
arams":["id":1]}' || echo "No response / timeout"; echo ""; done < enodes.txt  
==== RPC Test 157.173.127.30:8545 =====  
No response / timeout  
  
==== RPC Test 156.67.104.212:8545 =====  
No response / timeout  
  
==== RPC Test 144.91.74.181:8545 =====  
No response / timeout  
  
==== RPC Test 157.173.127.21:8545 =====  
No response / timeout  
  
==== RPC Test 157.173.127.31:8545 =====  
No response / timeout  
  
==== RPC Test 206.189.127.204:8545 =====  
{"jsonrpc":"2.0","id":1,"result":"0xe1749e"}  
  
==== RPC Test 161.97.89.27:8545 =====  
No response / timeout  
  
jeruzzalet@EdLWEISS:~$ curl -X POST http://IP:8545 ...  
curl: (6) Could not resolve host: IP  
curl: (6) Could not resolve host: ...  
jeruzzalet@EdLWEISS:~$ |
```

Full terminal output and additional test artifacts are available upon request.